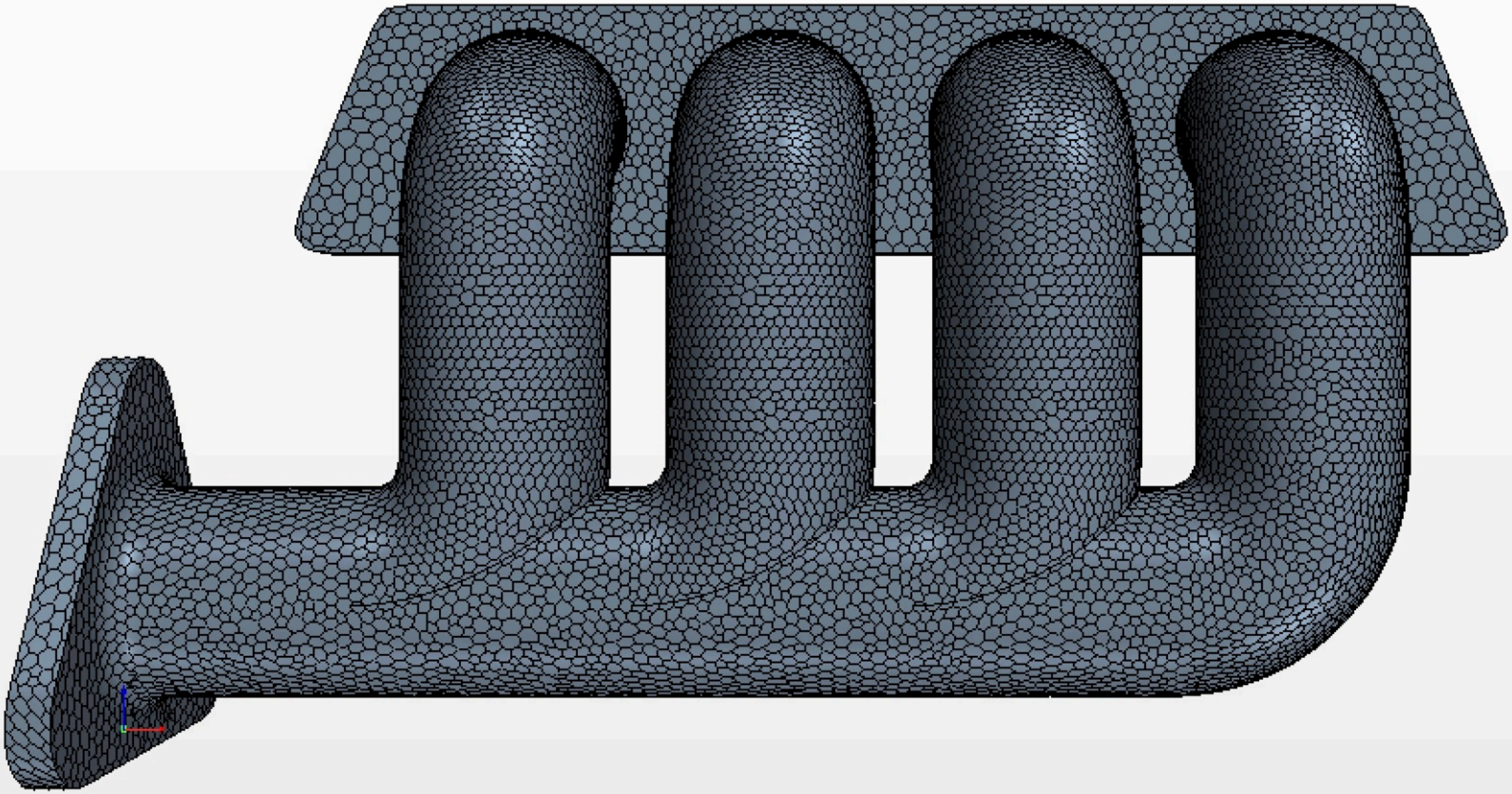
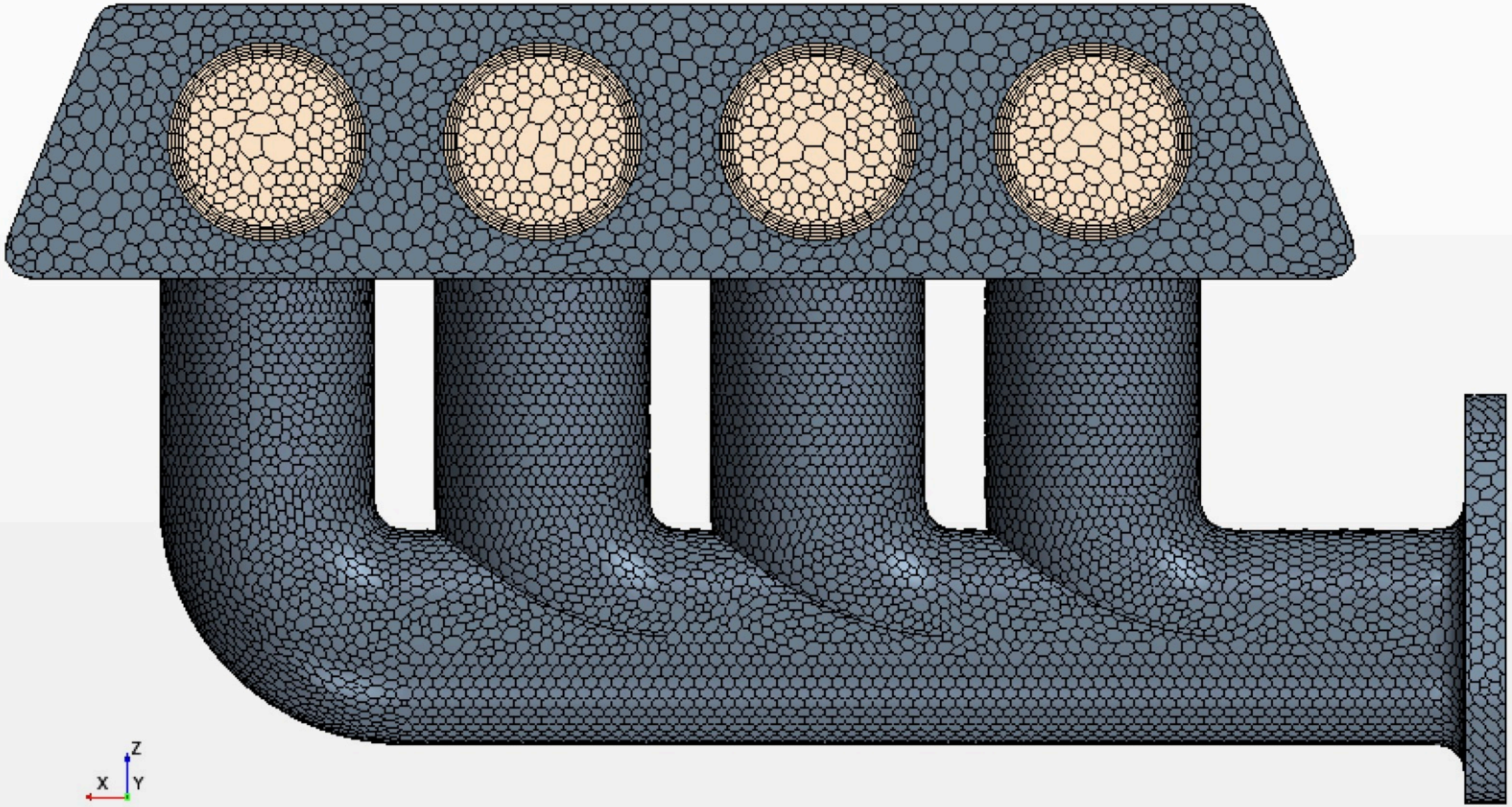
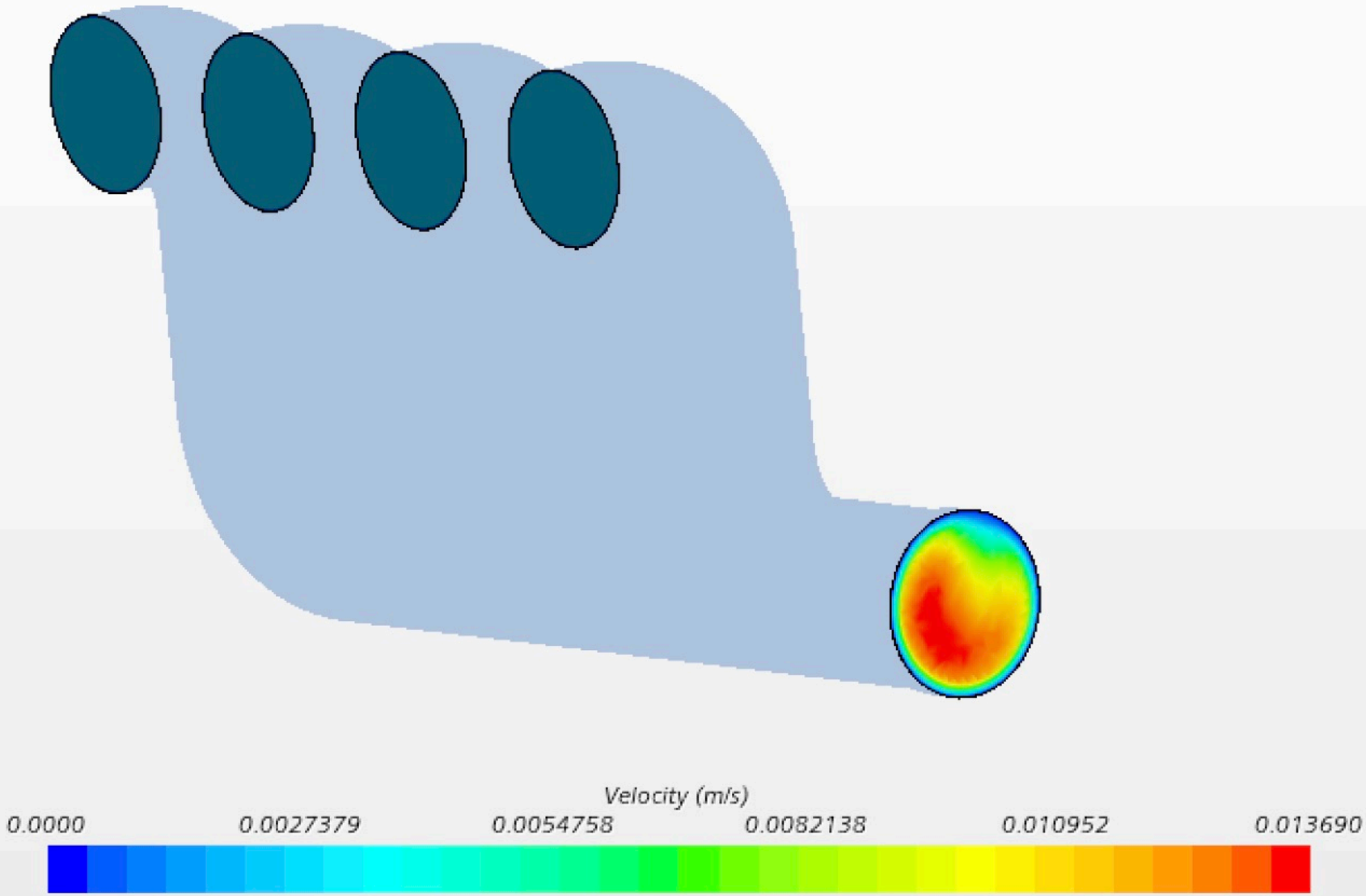
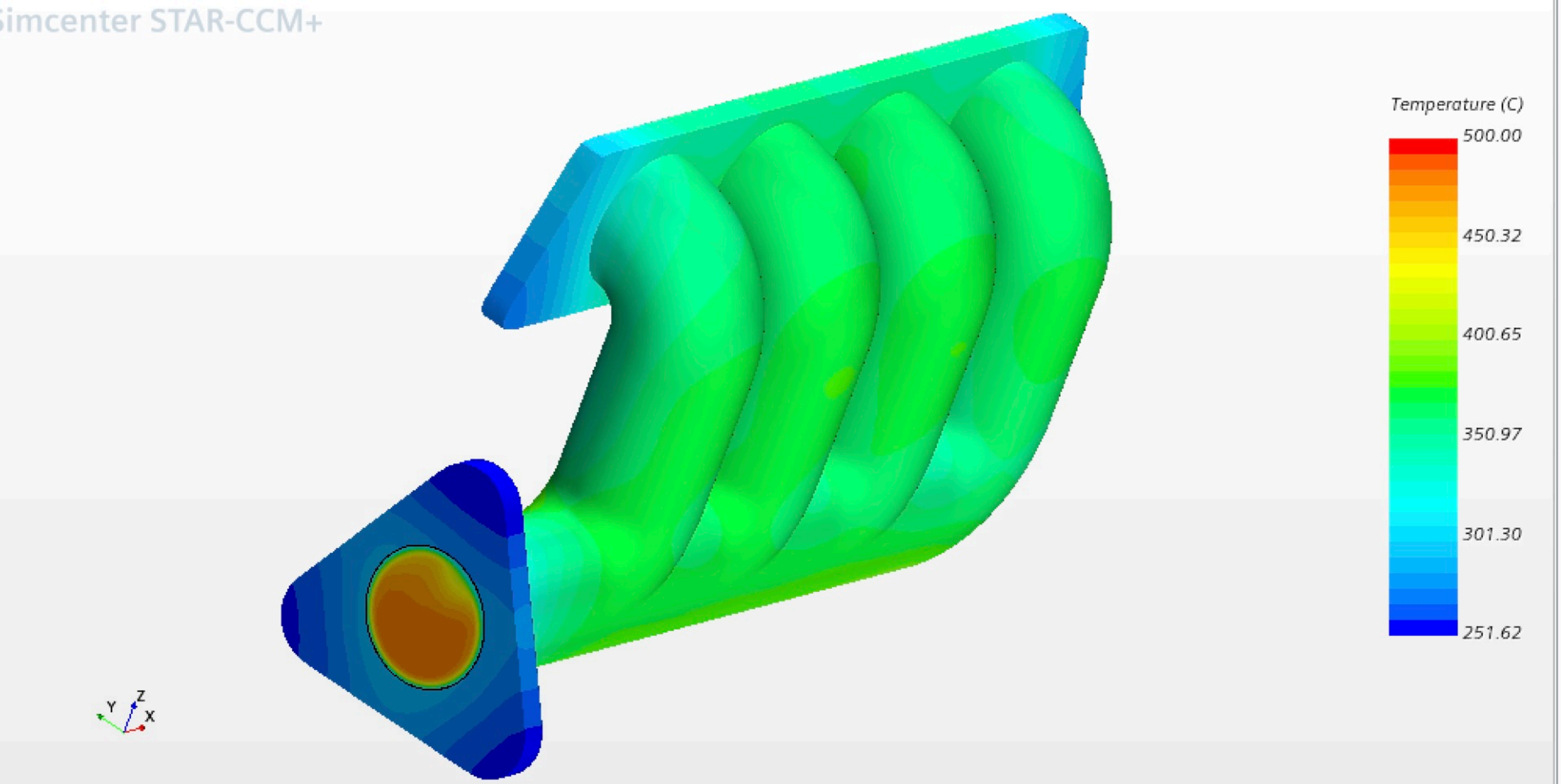
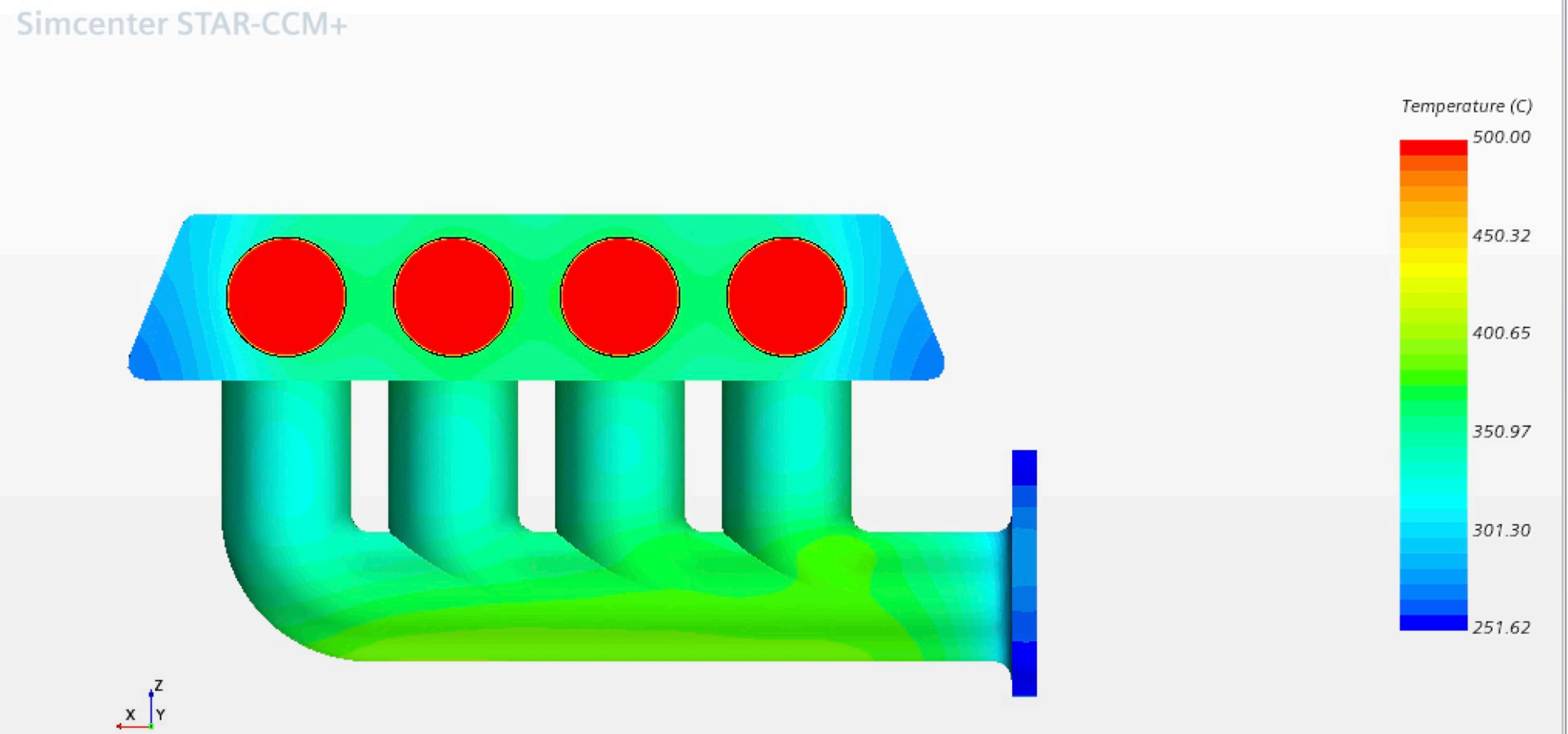
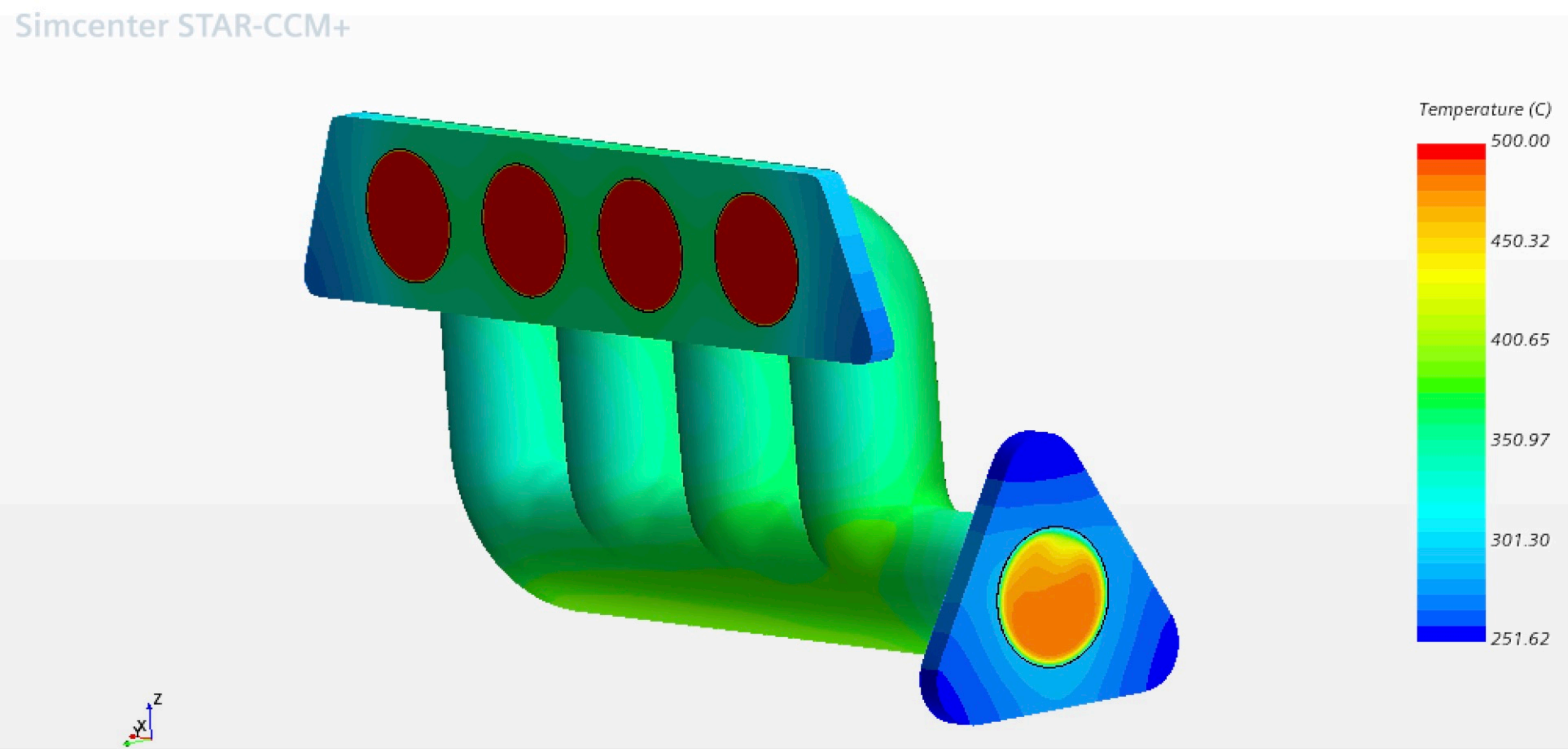



Z
X
Y


Velocity (m/s)
0.0000
0.0027379
0.0054758
0.0082138
0.010952
0.013690


Simcenter STAR-CCM+
Temperature (C)
500.00
450.32
400.65
350.97
301.30
251.62
Y
Z
X


Simcenter STAR-CCM+
Temperature (C)
500.00
450.32
400.65
350.97
301.30
251.62
Z
X
Y


Simcenter STAR-CCM+
Temperature (C)
500.00
450.32
400.65
350.97
301.30
251.62
Z
X

Summary Report: Manifold

Session Summary

Date	Feb 22, 2026, 11:43:34 AM
Simulation	Z:\Star CCM +\Manifold.sim
File size	94 MB
Number of Partitions	2
Number of Restored Partitions	1

Software Summary

Version	BuildArch: win64 BuildEnv: intel18.3vc14-r8 PresentationVersion: 2020.2.1 ReleaseDate: Thu Jul 16 15:33:18 UTC 2020 ReleaseNumber: 15.04.010
MPI Version	IBM Platform MPI-9.1.4.4

Hardware Summary

Hosts	Number Processes: 2 Rank[0]: VAMSHIKRISH8130 Rank[1]: VAMSHIKRISH8130

Simulation Properties

Manifold

Continua

Metal

Continua	3
Regions	[Region]
Interfaces	[Interface 1]
Point Sets	[]
Active	true
Tags	[]

Models

Constant Density

Gradients

Limiter Method	Venkatakrishnan
Custom Accuracy Level Selector	2.0
Verbose	false
Least-Squares Quality Criterion	true
Flat Cells Curvature Criterion	true
Cell Skewness Criterion	true
Chevron-Cell Criterion	true
Least-Squares Tensor Minimum Eigenvalues Ratio	0.1
Normalized Flat Cells Curvature Factor	1.0
Maximum Safe (Positive) Skewness Angle (deg)	75.0
Minimum Unsafe (Positive) Skewness Angle (deg)	88.0
Use TVB Gradient Limiting	false
Acceptable Field Variation (Factor)	0.05
Secondary Gradients	On

Segregated Solid Energy

Solid

UNSG101000

Database Material	UNSG101000 (Carbon Steel) [Standard/Solids]
Tags	[]

Material Properties

Density	Method	Constant
Constant	Value	7832.0 kg/m^3
Specific Heat	Method	Constant
Constant	Value	434.0 J/kg-K
Thermal Conductivity	Method	Constant
Constant	Value	63.9 W/m-K
Steady	Continuum Iteration	126
Three Dimensional		
Reference Values		
Minimum Allowable Temperature	Value	100.0 K
Maximum Allowable Temperature	Value	5000.0 K
Initial Conditions		
Static Temperature	Method	Constant
Constant	Value	90.0 C
Parts Meshes	Regions	[Region, Region 2]
	Interfaces	[Interface 1]
	Tags	[]
water	Regions	[Region 2]
	Interfaces	[Interface 1]
	Point Sets	[]
	Active	true
	Tags	[]
Models		
Constant Density		
Gradients	Limiter Method	Venkatakrishnan
	Custom Accuracy Level Selector	2.0
	Verbose	false
	Least-Squares Quality Criterion	true
	Flat Cells Curvature Criterion	true
	Cell Skewness Criterion	true
	Chevron-Cell Criterion	true
	Least-Squares Tensor Minimum Eigenvalues Ratio	0.1
	Normalized Flat Cells Curvature Factor	1.0
	Maximum Safe (Positive) Skewness Angle (deg)	75.0
	Minimum Unsafe (Positive) Skewness Angle (deg)	88.0
	Use TVB Gradient Limiting	false
	Acceptable Field Variation (Factor)	0.05
Gravity		
K-Epsilon Turbulence		
Liquid		
H2O	Database Material	H2O (Water) [Standard/Liquids]
	Tags	[]
Material Properties		
Density	Method	Constant

Constant	Value	997.561 kg/m^3
Dynamic Viscosity	Method	Constant
Constant	Value	8.8871E-4 Pa-s
Specific Heat	Method	Constant
Constant	Value	4181.72 J/kg-K
Thermal Conductivity	Method	Constant
Constant	Value	0.620271 W/m-K
Turbulent Prandtl Number	Method	Constant
Constant	Value	0.9
Realizable K-Epsilon Two-Layer	Buoyancy Production of Dissipation	Boundary Layer Orientation
	Cmu	0.09
	C1e	1.44
	C2e	1.9
	Ct	1.0
	Sigma_k	1.0
	Sigma_e	1.2
	Sarkar	2.0
	Tke Minimum	1.0E-10
	Tdr Minimum	1.0E-10
	Secondary Gradients	On
	Convection	2nd-order
	Normal Stress Term	false
	Curvature Correction Option	Off
	Two-Layer Type	Shear Driven (Wolfstein)
	Two-Layer ReY*	60.0
	Two-Layer Delta ReY	10.0
Reynolds-Averaged Navier-Stokes		
Segregated Flow	Positivity Rate Limit	0.2
	Flow Boundary Diffusion	true
	Unsteady Flux Dissipation Corrections	false
	Limit Acoustic-CFL Option	Per-Model
	Secondary Gradients	On
	Convection	2nd-order
	Delta-V Dissipation	Off
Segregated Fluid Temperature	Secondary Gradients	On
	Convection	2nd-order
	Flow Boundary Diffusion	true
Steady	Continuum Iteration	126
Three Dimensional		
Turbulent		
Two-Layer All y+ Wall Treatment	Iterative Ustar	false
Wall Distance	Wall Distance Method	Implicit Tree
Reference Values		

Gravity	Value	[0.0, 0.0, -9.81] m/s^2
Reference Altitude	Value	[0.0, 0.0, 0.0] m
Minimum Allowable Wall Distance	Value	1.0E-6 m
Maximum Allowable Absolute Pressure	Value	1.0E8 Pa
Minimum Allowable Temperature	Value	100.0 K
Minimum Allowable Absolute Pressure	Value	1000.0 Pa
Maximum Allowable Temperature	Value	5000.0 K
Reference Pressure	Value	101325.0 Pa
Initial Conditions		
Pressure	Method	Constant
Constant	Value	0.0 Pa
Static Temperature	Method	Constant
Constant	Value	400.0 C
Turbulence Intensity	Method	Constant
Constant	Value	0.01
Turbulence Specification	Method	Intensity + Viscosity Ratio
Turbulent Velocity Scale	Method	Constant
Constant	Value	30.0 m/s
Turbulent Viscosity Ratio	Method	Constant
Constant	Value	10.0
Velocity	Method	Constant
	Coordinate System	Laboratory
Constant	Value	[0.0, 0.0, 0.0] m/s
Interfaces	Verbosity	false
	Contact Selection Priority	[Interface 1]
	Overset Hierarchy	[]
	Interfaces	1
Interface 1	Allow Per-Contact Values	false
	Geometry	Boundaries
	Boundary-0	Region: Metal.Metal shell
	Boundary-1	Region 2: Water.Shell
	Contacts	[]
	Type	Contact Interface
	Topology	In-place
	Connectivity	Imprinted
	Tags	[]
Physics Conditions		
Energy Source Option	Energy Source Option	None
Thermal Specification	Condition	Conjugate Heat Transfer
Physics Values		
Contact Resistance	Method	Constant
Constant	Value	0.0 m^2-K/W
Intersection	Specify by Part Subgroup	false
Intersection tolerance	Value	0.05
Regions	Part Selection Priority	[Region, Region 2]

Region	Regions	2
	Index	0
	Mesh Continuum	Parts Meshes
	Physics Continuum	Metal
	Parts	[Metal]
	Type	Solid Region
	Allow Per-Part Values	false
	Tags	[]
Boundaries	Part Surface Selection Priority	[Region: Metal.Metal body, Region: Metal.Metal shell, Region: Metal.Metal shell [Interface 1]]
	Boundaries	3
Metal.Metal body	Index	2
	Interfaces	
	Part Surfaces	[Metal.Metal body]
	Type	Wall
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Thermal Specification	Condition	Convection
User Wall Heat Flux Coefficient Specification	Method	None
Physics Values		
Ambient Temperature	Method	Constant
Constant	Value	75.0 C
Heat Transfer Coefficient	Method	Constant
Constant	Value	50.0 W/m^2-K
Metal.Metal shell	Index	3
	Interfaces	Interface 1
	Part Surfaces	[Metal.Metal shell]
	Type	Wall
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Thermal Specification	Condition	Convection
User Wall Heat Flux Coefficient Specification	Method	None
Physics Values		
Ambient Temperature	Method	Constant
Constant	Value	75.0 C
Heat Transfer Coefficient	Method	Constant
Constant	Value	50.0 W/m^2-K
Metal.Metal shell [Interface 1]	Index	15
	Interfaces	

	Part Surfaces	[]
	Type	Contact Interface Boundary
	Allow Per-Contact Values	false
	Parent Interface	Interface 1
	Tags	[]
Physics Conditions		
User Wall Heat Flux Coefficient Specification	Method	None
Feature Curves	Part Curve Selection Priority	[Region:Feature Curve]
	Feature Curves	1
Feature Curve	Part Curves	[Metal.Default]
	Tags	[]
Physics Conditions		
Convective Velocity Option	Option	None
Energy Source Option	Energy Source Option	None
Initial Condition Option	Option	Use Continuum Values
Physics Values		
Axis	Coordinate System	Laboratory
	Origin	[0.0, 0.0, 0.0] m
	Direction	[0.0, 0.0, 1.0]
Motion Specification	Motion	Stationary
	Reference Frame	Lab Reference Frame
Region 2	Index	1
	Mesh Continuum	Parts Meshes
	Physics Continuum	water
	Parts	[Water]
	Type	Fluid Region
	Allow Per-Part Values	false
	Tags	[]
Boundaries	Part Surface Selection Priority	[Region 2: Water.Shell, Region 2: Water.Inlet 1, Region 2: Water.Inlet 2, Region 2: Water.Inlet 3, Region 2: Water.Inlet 4, Region 2: Water.outlet, Region 2: Water.Shell [Interface 1]]
	Boundaries	7
Water.Inlet 1	Index	6
	Interfaces	
	Part Surfaces	[Water.Inlet 1]
	Type	Mass Flow Inlet
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Flow Direction Specification	Method	Boundary-Normal
Mass Flow Option	Specification Option	Mass Flow Rate
Reference Frame Specification	Option	Lab Frame
Turbulence Specification	Method	Intensity + Viscosity Ratio
Physics		

Values		
Mass Flow Rate	Method	Constant
Constant	Value	0.01 kg/s
Total Temperature	Method	Constant
Constant	Value	500.0 C
Turbulence Intensity	Method	Constant
Constant	Value	1.0
Turbulent Viscosity Ratio	Method	Constant
Constant	Value	10.0
Water.Inlet 2	Index	7
	Interfaces	
	Part Surfaces	[Water.Inlet 2]
	Type	Mass Flow Inlet
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Flow Direction Specification	Method	Boundary-Normal
Mass Flow Option	Specification Option	Mass Flow Rate
Reference Frame Specification	Option	Lab Frame
Turbulence Specification	Method	Intensity + Viscosity Ratio
Physics Values		
Mass Flow Rate	Method	Constant
Constant	Value	0.01 kg/s
Total Temperature	Method	Constant
Constant	Value	500.0 C
Turbulence Intensity	Method	Constant
Constant	Value	1.0
Turbulent Viscosity Ratio	Method	Constant
Constant	Value	10.0
Water.Inlet 3	Index	8
	Interfaces	
	Part Surfaces	[Water.Inlet 3]
	Type	Mass Flow Inlet
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		

Flow Direction Specification	Method	Boundary-Normal
Mass Flow Option	Specification Option	Mass Flow Rate
Reference Frame Specification	Option	Lab Frame
Turbulence Specification	Method	Intensity + Viscosity Ratio
Physics Values		
Mass Flow Rate	Method	Constant
Constant	Value	0.01 kg/s
Total Temperature	Method	Constant
Constant	Value	500.0 C
Turbulence Intensity	Method	Constant
Constant	Value	1.0
Turbulent Viscosity Ratio	Method	Constant
Constant	Value	10.0
Water.Inlet 4	Index	9
	Interfaces	
	Part Surfaces	[Water.Inlet 4]
	Type	Mass Flow Inlet
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Flow Direction Specification	Method	Boundary-Normal
Mass Flow Option	Specification Option	Mass Flow Rate
Reference Frame Specification	Option	Lab Frame
Turbulence Specification	Method	Intensity + Viscosity Ratio
Physics Values		
Mass Flow Rate	Method	Constant
Constant	Value	0.01 kg/s
Total Temperature	Method	Constant
Constant	Value	500.0 C
Turbulence Intensity	Method	Constant
Constant	Value	1.0
Turbulent Viscosity Ratio	Method	Constant

Constant	Value	10.0
Water.outlet	Index	10
	Interfaces	
	Part Surfaces	[Water.outlet]
	Type	Pressure Outlet
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Backflow Specification	Direction	Boundary-Normal
	Pressure	Environmental
	Scalars	Specified
Pressure Outlet Option	Option	None
Reference Frame Specification	Option	Lab Frame
Turbulence Specification	Method	Intensity + Viscosity Ratio
Physics Values		
Pressure	Method	Constant
Constant	Value	0.0 Pa
Static Temperature	Method	Constant
Constant	Value	400.0 C
Turbulence Intensity	Method	Constant
Constant	Value	0.01
Turbulent Viscosity Ratio	Method	Constant
Constant	Value	10.0
Water.Shell	Index	5
	Interfaces	Interface 1
	Part Surfaces	[Water.Shell]
	Type	Wall
	Allow Per-Surface Values	false
	Tags	[]
Physics Conditions		
Reference Frame Specification	Option	Region Reference Frame
Shear Stress Specification	Method	No-Slip
Tangential Velocity Specification	Method	Fixed
Thermal Specification	Condition	Convection
User Wall Heat Flux Coefficient Specification	Method	None
Wall	Method	Smooth

Surface Specification		
Physics Values		
Ambient Temperature	Method	Constant
Constant	Value	400.0 C
Blended Wall Function	E	9.0
	Kappa	0.42
Heat Transfer Coefficient	Method	Constant
Constant	Value	50.0 W/m^2-K
Water.Shell [Interface 1]	Index	16
	Interfaces	
	Part Surfaces	[]
	Type	Contact Interface Boundary
	Allow Per-Contact Values	false
	Parent Interface	Interface 1
	Tags	[]
Physics Conditions		
Reference Frame Specification	Option	Region Reference Frame
Shear Stress Specification	Method	No-Slip
Tangential Velocity Specification	Method	Fixed
User Wall Heat Flux Coefficient Specification	Method	None
Wall Surface Specification	Method	Smooth
Physics Values		
Blended Wall Function	E	9.0
	Kappa	0.42
Feature Curves	Part Curve Selection Priority	[Region 2:Feature Curve]
	Feature Curves	1
Feature Curve	Part Curves	[Water.Default]
	Tags	[]
Physics Conditions		
Energy Source Option	Energy Source Option	None
Initial Condition Option	Option	Use Continuum Values
Mass Source Option	Mass Source Option	false
Momentum Source Option	Momentum Source Option	None
Turbulence Source Option	Turbulence Source Option	None
Physics Values		
Axis	Coordinate System	Laboratory
	Origin	[0.0, 0.0, 0.0] m
	Direction	[0.0, 0.0, 1.0]
Motion	Motion	Stationary

Specification		
	Reference Frame	Lab Reference Frame
Representations		
Latest Surface/Volume	Representation	Volume Mesh
	Tags	[]
Geometry	Tags	[]
Automated Mesh.Remesh	Tags	[]
Automated Mesh 2.Remesh	Tags	[]
Latest Surface	Tags	[]
Volume Mesh	Cells	107264
	Interior Faces	506195
	Vertices	476478
	Tags	[]
Finite Volume Regions		
Region	Cells	49624
	Interior Faces	234787
	Vertices	240634
	Edges	0
Finite Volume Boundaries		
Metal.Metal body	Faces	19061
Metal.Metal shell	Faces	0
Metal.Metal shell [Interface 1]	Faces	48842
Region 2	Cells	57640
	Interior Faces	271408
	Vertices	235844
	Edges	0
Finite Volume Boundaries		
Water.Inlet 1	Faces	268
Water.Inlet 2	Faces	267
Water.Inlet 3	Faces	264
Water.Inlet 4	Faces	295
Water.outlet	Faces	279
Water.Shell	Faces	0
Water.Shell [Interface 1]	Faces	48842
Cell Sets		
Contacts		
Water/Metal	Part 1	[Water]
	Part 2	[Metal]
Shell/Metal shell	Part Surface 1	Water.Shell
	Part Surface 2	Metal.Metal shell
	Conformality	STRONG_CONTACT
	Metadata	{}
	Index	2
	Interface	[]

	Tags	[]
Parts		
Metal	Metadata	{}
	Index	2
	Region	[Region]
	Contacts	[Water]
	Descriptions	[Root, Automated Mesh.Remesh]
	Face Count	151994
	Tags	[]
Surfaces		
Metal body	Index	2
	Metadata	{}
	Boundary	[Region: Metal.Metal body]
	Tags	[]
Metal shell	Index	8
	Metadata	{}
	Boundary	[Region: Metal.Metal shell]
	Tags	[]
Curves		
Default	Index	2
	Feature Curve	[Region:Feature Curve]
	Tags	[]
Water	Metadata	{}
	Index	1
	Region	[Region 2]
	Contacts	[Metal]
	Descriptions	[Root, Automated Mesh 2.Remesh]
	Face Count	76422
	Tags	[]
Surfaces		
Inlet 1	Index	3
	Metadata	{}
	Boundary	[Region 2: Water.Inlet 1]
	Tags	[]
Inlet 2	Index	4
	Metadata	{}
	Boundary	[Region 2: Water.Inlet 2]
	Tags	[]
Inlet 3	Index	5
	Metadata	{}
	Boundary	[Region 2: Water.Inlet 3]
	Tags	[]
Inlet 4	Index	6
	Metadata	{}
	Boundary	[Region 2: Water.Inlet 4]
	Tags	[]
outlet	Index	7
	Metadata	{}
	Boundary	[Region 2: Water.outlet]
	Tags	[]
Shell	Index	1
	Metadata	{}
	Boundary	[Region 2: Water.Shell]
	Tags	[]
Curves		
Default	Index	1
	Feature Curve	[Region 2:Feature Curve]
	Tags	[]
3D-CAD Models		
3D-CAD Model 1	Part Update Method	UPDATE_GEOMETRY

	Tags	[]
Body Groups		
exhaust manifold	Name	exhaust manifold
	Color	java.awt.Color[r=128,g=128,b=128]
	Opacity	1.0
	Override Upstream Color	false
	Tags	[]
Body 1	Name	Body 1
	Color	java.awt.Color[r=128,g=128,b=128]
	Opacity	1.0
	Tags	[]
Body 2	Name	Body 2
	Color	java.awt.Color[r=128,g=128,b=128]
	Opacity	1.0
	Tags	[]
Features		
XY	Error Message	
	Origin	[0.0, 0.0, 0.0] m
	X-Axis	[1.0, 0.0, 0.0]
	Y-Axis	[0.0, 1.0, 0.0]
	Tags	[]
YZ	Error Message	
	Origin	[0.0, 0.0, 0.0] m
	X-Axis	[0.0, 1.0, 0.0]
	Y-Axis	[0.0, 0.0, 1.0]
	Tags	[]
ZX	Error Message	
	Origin	[0.0, 0.0, 0.0] m
	X-Axis	[0.0, 0.0, 1.0]
	Y-Axis	[1.0, 0.0, 0.0]
	Tags	[]
Global Origin	Error Message	
	Position	[0.0, 0.0, 0.0]
	Tags	[]
Lab Coordinate System	Error Message	
	Origin	[0.0, 0.0, 0.0]
	X-axis Direction	[1.0, 0.0, 0.0]
	Y-axis Direction	[0.0, 1.0, 0.0]
	Tags	[]
ImportCad 1	Error Message	
	Exchange Settings	{STEP=0, NX=0, CATIAV5=0, SE=0, JT=0}
	File Name	Z:\VIVO\exhaust manifold.stp
	Tags	[]
Design Filters		
Design Parameters		
Operations		
Automated Mesh	Per-Part Meshing	false
	Mesher Execution Mode	Serial
	Input Parts	[Metal]
	Perform Local Surface Meshing	false
	Preserve Surface Perimeters	None
	Verbose Output	false
	Tags	[]
Meshers		
Surface Remesher	Meshing Method	Triangle
	Perform Curvature Refinement	true

		Perform Proximity Refinement	true
		Perform Compatibility Refinement	false
		Create Aligned Meshes	true
		Minimum Face Quality	0.05
Mesher	Polyhedral	Run Post Mesh Optimizer	true
Mesher	Prism Layer	Stretching Function	Geometric Progression
		Distribution Mode	Stretch Factor
		Boundary March Angle	50.0
		Gap Fill Percentage	25.0
		Minimum Thickness Percentage	10.0
		Layer Reduction Percentage	50.0
		Concave Angle Limit	0.0
		Convex Angle Limit	360.0
		Near Core Layer Aspect Ratio	0.0
	Default Controls		
	Base Size	Base Size	0.05 m
		Base Size	0.05 m
Projection	CAD	Project to CAD	true
Surface Size	Target	Size Type	Relative to base
		Percentage of Base	50.0
		Absolute Size	0.025 m
Surface Size	Minimum	Size Type	Relative to base
		Percentage of Base	10.0
		Absolute Size	0.005 m
Curvature	Surface	Enable Curvature Deviation Distance	false
		# Pts/circle	36.0
		Max # Pts/circle	200.0
		Curvature Deviation Distance	0.01 m
Proximity	Surface	Search Floor	0.0 m
		# Points in gap	2.0
		Enable Search Ceiling	false
		Search Ceiling	1.0E10 m
Growth Rate	Surface	Surface Growth Rate	SLOW
		User Specified Value	1.3
Prism Layers	Number of	Number of Prism Layers	6
Stretching	Prism Layer	Prism Layer Stretching	1.0
Total Thickness	Prism Layer	Size Type	Relative to base
		Percentage of Base	25.0
		Absolute Size	0.0125 m
Growth Rate	Volume	Volume Growth Rate	1.2
Tet Size	Maximum	Size Type	Relative to base
		Percentage of Base	10000.0
		Absolute Size	5.0 m
Optimization	Core Mesh	Optimization Cycles	1
		Quality Threshold	0.4
Optimization	Post Mesh	Optimize Boundary Vertices	false

		Optimize Cell Topology	false
	Custom Controls		
2	**Automated Mesh**	Per-Part Meshing	false
		Mesher Execution Mode	Serial
		Input Parts	[Water]
		Perform Local Surface Meshing	false
		Preserve Surface Perimeters	None
		Verbose Output	false
		Tags	[]
	Meshers		
	Surface Remesher	Meshing Method	Triangle
		Perform Curvature Refinement	true
		Perform Proximity Refinement	true
		Perform Compatibility Refinement	false
		Create Aligned Meshes	true
		Minimum Face Quality	0.05
	Polyhedral Mesher	Run Post Mesh Optimizer	true
	Prism Layer Mesher	Stretching Function	Geometric Progression
		Distribution Mode	Stretch Factor
		Boundary March Angle	50.0
		Gap Fill Percentage	25.0
		Minimum Thickness Percentage	10.0
		Layer Reduction Percentage	50.0
		Concave Angle Limit	0.0
		Convex Angle Limit	360.0
		Near Core Layer Aspect Ratio	0.0
	Default Controls		
	Base Size	Base Size	0.05 m
		Base Size	0.05 m
	CAD Projection	Project to CAD	true
	Target Surface Size	Size Type	Relative to base
		Percentage of Base	50.0
		Absolute Size	0.025 m
	Minimum Surface Size	Size Type	Relative to base
		Percentage of Base	10.0
		Absolute Size	0.005 m
	Surface Curvature	Enable Curvature Deviation Distance	false
		# Pts/circle	36.0
		Max # Pts/circle	200.0
		Curvature Deviation Distance	0.01 m
	Surface Proximity	Search Floor	0.0 m
		# Points in gap	2.0
		Enable Search Ceiling	false
		Search Ceiling	1.0E10 m
	Surface Growth Rate	Surface Growth Rate	DEFAULT
		User Specified Value	1.3
	Number of Prism Layers	Number of Prism Layers	4
	Prism Layer Stretching	Prism Layer Stretching	1.0
	Prism Layer	Size Type	Relative to base

Total Thickness		
	Percentage of Base	10.0
	Absolute Size	0.005 m
Volume Growth Rate	Volume Growth Rate	1.2
Maximum Tet Size	Size Type	Relative to base
	Percentage of Base	10000.0
	Absolute Size	5.0 m
Core Mesh Optimization	Optimization Cycles	1
	Quality Threshold	0.4
Post Mesh Optimization	Optimize Boundary Vertices	false
	Optimize Cell Topology	false
Custom Controls		
Descriptions	Number of Children	5
Root	Described Parts	[Water, Metal]
Automated Mesh.Remesh	Described Parts	[Metal]
	Faces	66400
	Vertices	33188
Automated Mesh 2.Remesh	Described Parts	[Water]
	Faces	19078
	Vertices	9541
Latest Surface	Described Parts	[Water, Metal]
	Faces	85478
	Vertices	42729
	Preview Mesh Operation Parts	false
Latest Surface/Volume	Described Parts	[Water, Metal]
Coordinate Systems		
Laboratory	Tags	[]
Local Coordinate Systems		
Tables	Tables	0
Tags		
Units	Preferred System	Systeme Internationale
Custom Trees		
Filters		
Field Functions		
Simulation Operations	Selected	[]
Parameters		
Volume Shapes		
Data Set Functions	Data Directory	function_data
Update Events	Event Count	0
	Event Names	
Time Scales		
User Code		
Data Focus		
Layouts		
default		
Data Mappers	Use Legacy Volume Mapper	false
Motions		
Stationary	Tags	[]
Reference Frames		
Lab Reference Frame	Tags	[]
Screenplays		
Derived Parts	Derived Parts	0

Summaries		
Monitors	Monitors	12
	Monitors To Print	[Energy, Continuity, X-momentum, Y-momentum, Z-momentum, Tke, Tdr, Average metal temperature Monitor, Average water temperature Monitor, Temperature at outlet Monitor]
	Output Direction	Horizontal
	Heading Print Frequency	10
Average metal temperature Monitor	Report	[Average metal temperature]
	Enabled	true
	Value Type	Total Value
	Trigger	Iteration
	Normalization Option	Off
	Maximum Plot Samples	5000
	Tags	[]
Iteration Frequency	Iteration Frequency	1
	Start Iteration	0
	Enable Stop	false
	Stop Iteration	0
Average water temperature Monitor	Report	[Average water temperature]
	Enabled	true
	Value Type	Total Value
	Trigger	Iteration
	Normalization Option	Off
	Maximum Plot Samples	5000
	Tags	[]
Iteration Frequency	Iteration Frequency	1
	Start Iteration	0
	Enable Stop	false
	Stop Iteration	0
Iteration	Maximum Plot Samples	5000
	Tags	[]
Physical Time	Maximum Plot Samples	5000
	Tags	[]
Temperature at outlet Monitor	Report	[Temperature at outlet]
	Enabled	true
	Value Type	Total Value
	Trigger	Iteration
	Normalization Option	Off
	Maximum Plot Samples	5000
	Tags	[]
Iteration Frequency	Iteration Frequency	1
	Start Iteration	0
	Enable Stop	false
	Stop Iteration	0
Reports	Reports	3
Average metal temperature	Units	C
	Field Function	Temperature
	Parts	[Region: Metal.Metal body, Region: Metal.Metal shell]
	Representation	Volume Mesh
	Smooth Values	false
	Signed Mass Flow	false

Average water temperature	Tags	[]
	Units	C
	Field Function	Temperature
	Parts	[Region 2: Water.Inlet 1, Region 2: Water.Inlet 2, Region 2: Water.Inlet 3, Region 2: Water.Inlet 4, Region 2: Water.outlet, Region 2: Water.Shell]
	Representation	Volume Mesh
	Smooth Values	false
	Signed Mass Flow	false
	Tags	[]
Temperature at outlet	Units	K
	Field Function	Temperature
	Parts	[Region 2: Water.outlet]
	Representation	Volume Mesh
	Smooth Values	false
	Signed Mass Flow	false
	Tags	[]
Solvers		
Steady		
Stopping Criteria	Verbose	false
Partitioning	Solver Frozen	false
	Partitioning Method	Per-Continuum
Wall Distance	Solver Frozen	false
	Verbosity	0
	Minimum Tree Size Threshold	500000
Segregated Flow	Pressure Gradient Inverse Distance Weighting	true
	Freeze Flow	false
	Reconstruction Frozen	false
	Reconstruction Zeroed	false
	Enable Enhanced Stability Treatment	false
	Temporary Storage Retained	false
	Pressure Corrections: Bad Cell Minimum Scaling	0.8
	Pressure Corrections: Acceptable Cell Volume Change	0.001
	Velocity Corrections: Maximum Unlimited Velocity	20.0 m/s
	Velocity Corrections: Acceptable Velocity Increase Rate [<1]	0.15
	Continuity Initialization	false
Velocity	Under-Relaxation Factor	0.7
	Dynamic Local Under-Relaxation	false
Under-Relaxation Factor Ramp	Ramp Method	No Ramp
AMG Linear Solver	Max Cycles	30
	Verbosity	NONE
	Enable Direct Solver	false
	Maximum Direct Solver Equations	32
	Convergence Tolerance	0.1
	Epsilon	0.0
	Cycle Type	Flex Cycle
	Group Size Control	Auto
	Group Size	4
	Relaxation Scheme	Gauss-Seidel
	Acceleration Method	None
	Scaling	Disabled

Flex Cycle	Restriction Tolerance	0.9
	Prolongation Tolerance	0.5
	Sweeps	1
Pressure	Number of Non-Orthogonality Correctors	0
	Under-Relaxation Factor	0.3
	Pressure Reference Location	Automatic Selection
Under-Relaxation Factor Ramp	Ramp Method	No Ramp
AMG Linear Solver	Max Cycles	30
	Verbosity	NONE
	Enable Direct Solver	false
	Maximum Direct Solver Equations	32
	Convergence Tolerance	0.1
	Epsilon	0.0
	Cycle Type	V Cycle
	Group Size Control	Auto
	Group Size	4
	Relaxation Scheme	Gauss-Seidel
	Acceleration Method	Conjugate Gradient
	Scaling	Auto
V Cycle	Pre-Sweeps	1
	Post-Sweeps	1
	Max Levels	50
Segregated Energy	Solver Frozen	false
	Reconstruction Frozen	false
	Reconstruction Zeroed	false
	Temporary Storage Retained	false
	Fluid Under-Relaxation Factor	0.9
	Solid Under-Relaxation Factor	0.99
	Enable High-Accuracy Temporal Discretization	false
Fluid Under-Relaxation Factor Ramp	Ramp Method	No Ramp
Solid Under-Relaxation Factor Ramp	Ramp Method	No Ramp
AMG Linear Solver	Max Cycles	30
	Verbosity	NONE
	Enable Direct Solver	false
	Maximum Direct Solver Equations	32
	Convergence Tolerance	0.1
	Epsilon	0.0
	Cycle Type	V Cycle
	Group Size Control	Auto
	Group Size	4
	Relaxation Scheme	Gauss-Seidel
	Acceleration Method	None
	Scaling	Disabled
V Cycle	Pre-Sweeps	1
	Post-Sweeps	1
	Max Levels	50
K-Epsilon Turbulence	Solver Frozen	false
	Reconstruction Frozen	false
	Reconstruction Zeroed	false
	Temporary Storage Retained	false
	Under-Relaxation Factor	0.8
	Boundary Layer Initialization	false
Under-	Ramp Method	No Ramp

Relaxation Factor Ramp		
AMG Linear Solver	Max Cycles	30
	Verbosity	NONE
	Enable Direct Solver	false
	Maximum Direct Solver Equations	32
	Convergence Tolerance	0.1
	Epsilon	0.0
	Cycle Type	Flex Cycle
	Group Size Control	Auto
	Group Size	4
	Relaxation Scheme	Gauss-Seidel
	Acceleration Method	None
	Scaling	Disabled
Flex Cycle	Restriction Tolerance	0.9
	Prolongation Tolerance	0.5
	Sweeps	1
K-Epsilon Turbulent Viscosity	Solver Frozen	false
	Under-Relaxation Factor	1.0
	Maximum Ratio	100000.0
Stopping Criteria	Verbose	false
Maximum Steps	Enabled	true
	Maximum Steps	200
	Logical Rule	Or
	Criterion Satisfied	false
	Tags	[]
Stop File	Enabled	true
	Stop Inner Iterations	true
	Path	ABORT
	Logical Rule	Or
	Criterion Satisfied	false
	Tags	[]
Solution Histories		
Solution Views		
Current Solution	Iteration	126
	Time Step	0
	Solution Time	0.0
	Tags	[]
Layout Views		

Solution

Accumulated CPU Time over all processes (s)	108.06700000000008
Elapsed Time (s)	108.06400000000006
Iterations	126